22BCON1060

ASSIGNMENT - 01

14/03/24

Sec - A

COD

PAGE NO.: 01

1. Convert the decimal number 53 to binary, octal, and hexadecimal representations.

Ans

Binary : $(53)_{10} = (110101)_2$

Octal : $(53)_{10} = (65)_8$

Hexadecimal : $(53)_{10} = (35)_{16}$

2. Define the term "bus" in the context of computer architecture and explain its types.

Ans A bus is a set of electrical wires (lines) that connects the various hardware components of a computer system. it works as a communication pathway through which information flow from one hardware component to the other hardware component. types of bus are :

(i) Data Bus : it is used for transmitting the data and it is bi-directional.

(ii) Control Bus : it is used to transfer the control and timing signals from one component to other.

(iii) Address Bus : it is used to carry address from CPU to memory / IO devices.

3. What is the role of the Arithmetic Logic Unit (ALU) in a CPU?

Ans ALU is a main component of the Central Processing unit and performs arithmetic and logic operations. it is also known as an integer unit (IU) that is an integrated circuit within a CPU or GPU, which is the last component to perform calculations in the processor. it has the ability to perform all process related to arithmetic and logic operations such as addition, subtraction, and shifting operations, including boolean comparisons (XOR, OR, AND, NOT).

Q4. Find R' compliment of
(i) $(11011)_2$
(ii) $(BAC)_{16}$
(iii) $(703)_8$

(i) $(11011)_2$
R' compliment of $(11011)_2 = (00100)_2$

(ii) $(BAC)_{16}$
R' compliment of $(BAC)_{16} = (453)_{16}$

(iii) $(703)_8$
R' compliment of $(703)_8 = (074)_8$

Q5. What are the purpose of a memory hierarchy in computer architecture? Briefly explain.

Ans Memory Hierarchy is about arranging different kinds of storage devices in a computer based on their size, cost and access speed, and the roles they play in application processing. The main purpose is to achieve efficient operations by organizing the memory to reduce access time while speeding up operations.

• Section-B (7x3=21)

1. Explain how to perform division using binary numbers. provide an example.

Ans The binary division is an important part of binary arithmatic and it is similar to that of a decimal division operation.

Rules of Binary division: There are four rules associated with binary division. The binary division rules are as follows:

- $1 \div 1 = 1$
- $1 \div 0 = 1$

- $0 \div 1$ = meaningless
- $0 \div 0$ = meaningless

As, binary numbers include only two digits i.e: 0 and 1 these four rules are all posible conditions for the division of binary numbers.

Four steps to Binary division:

(i) Division - First take the leftmost digit of the dividend, we attempt to divide it by divisor which must be smaller than the dividend digits. This resut in a quotient.

(ii) Multiplication - once we have found the quotient are we it to multipy the divisior to obtain a product.

(iii) subtraction - subtract that from the working dividend to calculate remainder.

(iv) Bring Down - The final step is then to bring down the next digit in our original dividend.

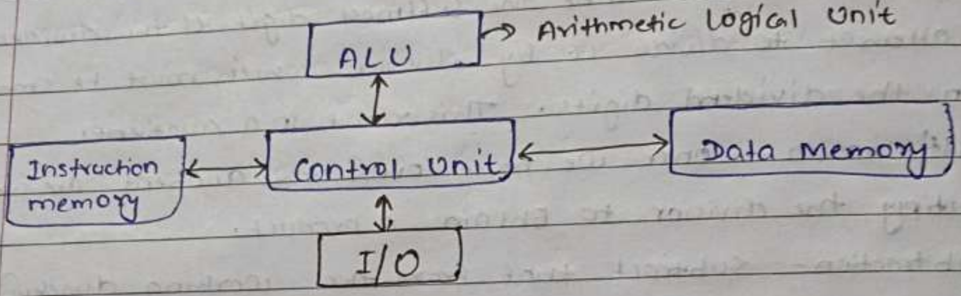
Example: Divide $0111100 \div 0010$

The zero is most significant bits does'not change value of number so remove the zero's. $111100 \div 10$

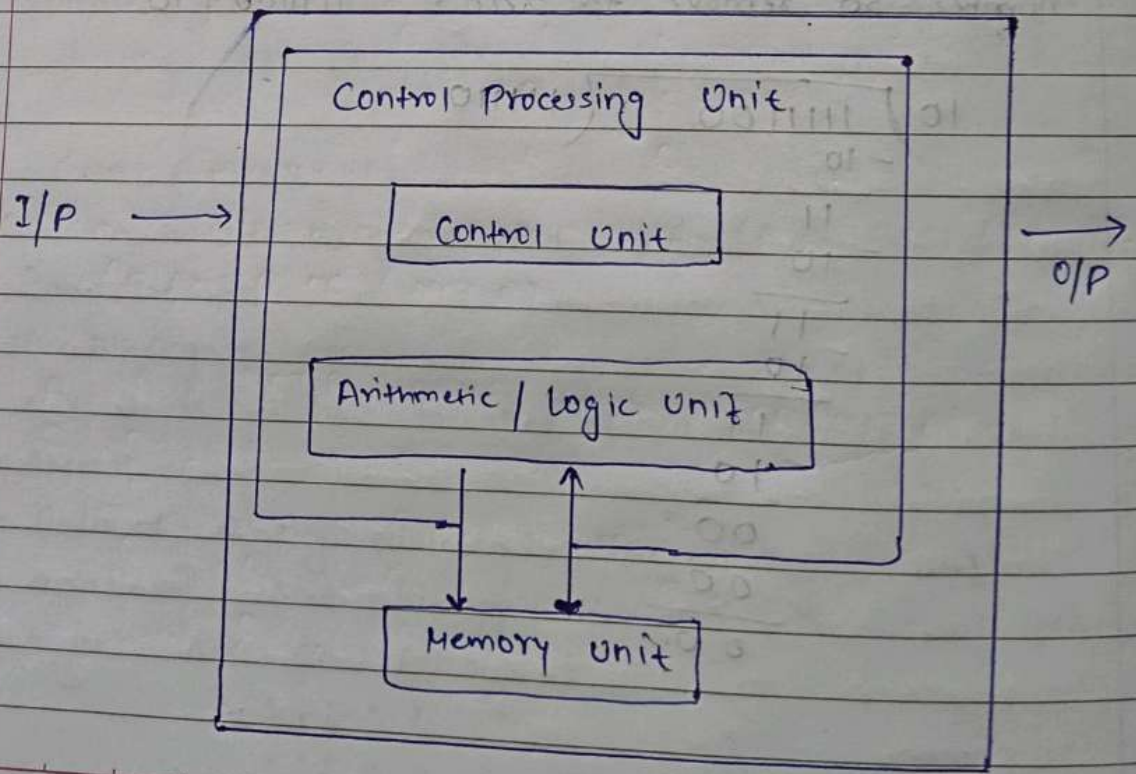
```
10 ) 111100 ( 11110
    -10
     11
    -10
     11
    -10
     11
    -10
     00
     00
     00
```

2. Compare and contrast Hardvaed arch. and Von Neumann arch.

Ans. Hardvard Architecture : it is the digital computer architecture whose design is based on the concept where there are seperate storage and seperate buses (single path) for instruction and data. it was basically developed to overcome the bottleneck of von Neumann architecture.



• Von Neumann Architecture : it is a digital computer architecture where design is based on the concept of stored program computers where program data and instruction data are stored in the same memory.



- Difference between Von Neumann & Hardvard Arch. :

Von Neumann Architecture	Hardvard Architecture
• it is cheaper.	• it is costly.
• CPU cannot exist instructions and read/write at the same time.	• CPU can exist instructions and read/write data at the same time.
• it is used in personal computers and small computers.	• it is used in microcontrolers and single processing.
• There is common bus for data and instruction transfer.	• seperate buses are used for transfering data and instructions.

Q3. (i) Define the basic logic gates (AND, OR, NOT, XOR) and explain their truth tables.

Ans (i) AND Gate : in the AND gate, the output of an AND gate attains state 1 if and only if all the inputs are in state 1. Boolean expression : $Y = A \cdot B$

Truth table :

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

- OR Gate : in an OR gate, the output of an OR gate attains state 1 if one or more input attains state 1.

Boolean expression : $Y = A + B$

Truth table :

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

- NOT Gate :- The output of a NOT gate attains state 1 if and only if the input does not attain state 1.

  Truth table :

A	Y
0	1
1	0

- XOR Gate : The output of a two-input XOR Gate attains state 1 if one adds only input attains state 1.

  Boolean expression : $A \cdot \bar{B} + \bar{A} \cdot B$ or $Y = A \oplus B$

  Truth table :

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Q (ii) Construct a truth table for the logical operation (A AND B) OR (NOT A).

Ans. (A AND B) OR (NOT A)

A	B	NOT A	A AND B	(A AND B) OR (NOT A)
0	0	1	0	1
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1

Section-C

Q1. Discuss the significance of signed-magnitude representation in computer arithmetic. Also Discuss the advantages of using floating-point representation over fixed-point representation in numerical computations.

Ans Signed-magnitude representation is a convention in which ~~one~~ express a decimal number as a positive or negative binary number using its most significant bit. The most significant bit is the left most bit, which represents the sign of a binary digit: if it is zero(0) then its positive, and if it is one(1) then it represents a negative number.

Ex: +9 is represented as 01001
-9 is represented as 11001

- The advantages of using floating-point representation over fixed point representation in numerical computation are:

(i) Flexibility in range and precision: Floating point numbers can represent a much wider range of values compared to fixed point number.

(ii) Efficient handling of large and small numbers: Floating point numbers are a dynamic range which means they can efficiently handle numbers that vary greatly in magnitude.

(iii) Ease of use in Mathematical Expression: Floating-point numbers allow for natural representation of mathematical expression as they can handle both integer and fractional parts without requiring explicit scaling or normalization.

(iv) Standardization: Floating-point representation is standardized by the IEEE 754 standard, which ensures consistency in representation and arithmatic operations across different platforms and programming Languages.

(v) Error Handling : Floating - point representation includes mechanisms for handling errors, such as rounding and overflow, which can help ensure that computations are more robust and reliable.

Overall, the flexibility, efficiency and standardized nature of floating-point representation make it a prefered choice for numerical computations in a wide range of applications.

Q2. What is the number system? Define all types and convert the flowing :

(i) $(110.1101)_2 = (?)_{16}$

(ii) $(256.51)_8 = (?)_{10}$

(iii) $(AE10)_{16} = (?)_2$

Ans Number System : A number system is a way to represent numbers using symbols or digits, along with rules for performing arithmetic operations such as addition subtraction, multiplication and division.

There are several types of Number system are :

(i) Binary Number System (Base 2) : Uses only two digit, 0 and 1. it is widely used in digital electronics and Computer Systems.

(ii) Decimal Number System (Base 10) : Use ten digits, 0 through 9, and is the most commonly used number system in arithmetic.

(iii) Octal Number System (Base 8) : uses digits 0 to 7. Octal numbers are often used in computing for representing

(iv)

group of bits, as they align neatly with binary digits.

(iv) Hexadecimal Number System: Uses digits 0 to 9 and letters A to F (representing values 10 to 15). Hexadecimal is commonly used for computing for its compact representation of binary data.

(i) $(1101101)_2 = (?)_{16}$

Group of binary digit into four sets of digits starting from right and convert each group into its hexadecimal equivalent.

$\Rightarrow (\underline{0110} \quad \underline{1101})_2$

0110 = 6 (Hexadecimal equivalent of 0110)

1101 = D (Hexadecimal equivalent of 1101)

So, $(1101101)_2 = (6D)_{16}$ Ans

(ii) $(256.51)_8 = (?)_{10}$

$256.51 = (2\times8^2) + (5\times8^1) + (6\times8^0) + (5\times8^{-1}) + (1\times8^{-2})$

$= 174.640625$

So, $(256.51)_8 = (174.640625)_{10}$ Ans

(iii) $(AE10)_{16} = (?)_2$

A	E	1	0
1010	1110	0001	0000

$\Rightarrow$ 1010111000010000

So $(AE10)_{16} = (1010111000010000)_2$

Ans

Q3. Perform binary Addition / subtraction on following :

(i) $(100110)_2 + (1101)_2$

$$\begin{array}{r} 100110 \\ +\ 1101 \\ \hline 110011 \end{array}$$

So, $(100110)_2 + (1101)_2 = (110011)_2$ A

(ii) $(11100)_2 - (1111)_2$

$$\begin{array}{r} 11100 \\ -\ 1111 \\ \hline 01101 \end{array}$$

So, $(11100)_2 - (1111)_2 = (1101)_2$ Ans

(iii) $(-78) + (-67)$

78 in binary = 01001110
67 in binary = 01000011

For (-78),

1's complement of 78, 10110001

2's complement,
$$\begin{array}{r} 10110001 \\ +\ \ 1 \\ \hline 10110010 \end{array}$$
i.e. -78

For (-67)

1's complement of 67, 10111100

2's complement,
$$\begin{array}{r} 10111100 \\ +\ \ 1 \\ \hline 10111101 \end{array}$$
i.e. -67

Now, (-78) + (-67)

```
   10110010
+  10111101
 1 01101111
```

Here, most significant digit 1 (left most) represent negative (-).

1's complement of 01101111 is 10010000

2's complement,
```
   10010000
+         1
   10010001
```
i.e. (-145)

So, -10010001 is Ans

x——x——x——x——x——